Anomaly Detection in IoT Networks Using Classical Machine Learning Techniques

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1 Introduction

The Internet of Things is a system of global physical objects, vehicles, sensing devices and actuating devices that communicate and share data through standard Internet protocols, without requiring direct human involvement to do so.

In the early 2020's, the number of connected devices is measured in tens of billions, with these devices positioned in almost every possible location across consumer households, manufacturing floor space, hospitals and transportation and critical public infrastructure, etc. [1, 8].The large scale at which we currently deploy technology has considerably altered both the definition of network security in the real world, as well as the nature of the attack surface available to adversary actors. The traditional perimeter-based model of defending against attacks is

quickly becoming outdated because the volume and variety of devices that join and compose these networks seem to be multiplying at rates that overwhelm the capacity of perimeter-based security mechanisms.

Anomaly Based Detection is one of the more principled ways to reflect to this fundamental change. Instead of relying on a list of known attack signatures against which we may identify and block malicious traffic, anomaly based detection systems first understand the normal behaviour of networks before determining and flagging any deviations from that normal behaviour. This approach is very relevant to the IoT context as a result of both the speed at which new attack types are coming into the world and the fact that most IoT devices operate in a consistent manner, making even small deviations from those consistent behaviours diagnostic signals. [3, 5].A temperature sensor that suddenly starts scanning subnets has an abnormality in terms of what it usually does, even if the specific way that the sensor is scanning has yet to make it into any databases.

Traditional machine learning approaches to IoT anomaly detection (i.e., those developed prior to the advent of deep learning) have been successfully used in practice. Support vector machines, random forests, isolation forests, k-nearest neighbour classifiers and density-based clustering methods all appear frequently in the published literature on

this topic.

These techniques are computationally cost-effective, more easily explained than neural networks, and appropriate for use on the edge and gateway devices typical of many IoT applications. [11, 13]. They are the focus of this chapter.

1.1 The Scale of IoT Deployment and Its Security Consequences

Out of all of these great devices that compose the Internet of Things, security was not thought about when these products were initially created, as a whole.

Many of these consumer IoT products are produced by hundreds of different manufacturers and the devices are built using different levels and types of quality standards. Then, we're faced with so many devices created using old firmware that have not been updated, while there are also devices created without a way for end users to receive security updates through either device updates or firmware updates.

In addition to the faulty security of most industrial IoT devices, there are additional issues related to the security of IoT devices due to the fact that operational technology networks/technologies have always been on a separate physical network from the internet. This was because they were never intended to be connected to an outside network and therefore do not use any form of cryptography to control access to these devices. [1].

In 2016, the world saw the real ramifications of this trend through the Mirai botnet, which used hundreds of thousands of hacked camera and home routers to carry out a massive distributed denial-of-service attack (DDoS) that crippled large portions of the Internet as we know it today. Koirala, Bista and Ferreira then went on to prove that when a consumer has had their Internet of Things (IoT) device attacked via a botnet, the attack can be detected and analysed through the use of machine learning classifiers trained on per-device traffic data and therefore presents a measurable change in the way a device behaves on a network after it has been hacked. [12]. There are more studies that support this data. A study conducted by Churcher illustrated the ability of a variety of supervised classification systems (like k-nearest neighbor, support vector machine, and random forest) to classify Bot-IoT attack traffic. This provides evidence that behavior can be indicated using fairly simple model systems. [4]. The anomaly detection framing is motivated by these considerations. A device owner cannot detect whether their device has been compromised through the observation of its network traffic if the device owner does not see what's happening. The security system that relies on the prior understanding of how to recognize attack activity will always be one step behind the attackers. Detection based on the understanding of learned normal behaviour solves both of these issues at the same time; on the condition that the models used in detection are of good quality and that the implementation of the models is feasible and practical for use in the environments in which IoT devices are deployed.

1.2 Why Classical Machine Learning Remains Practically Relevant

Since about 2015, deep learning algorithms have gained a lot of attention for their ability to perform anomaly detection. Convolutional neural networks (CNNs), recurrent neural networks (RNNs) and autoencoders have been used to assess network traffic and sometimes achieve stellar performance. Unfortunately, the increasing hype surrounding deep learning has, at times, clouded its real-world limitations.

Training these models requires a great deal of computer power, they need large amounts of labelled data, and the results can be difficult to interpret when an anomaly is detected. [15]. From an operational perspective, it can be challenging for security operators to communicate alerts or react quickly to incidents where there is no way of verifying the correctness or reliability of the black-box.

Not all classical ML techniques exhibit the same degree of weakness across all of these areas. equally [11]. An analyst can determine what traffic traits caused a specific classification by looking at a feature importance score generated by a Random Forest.

An anomaly score from an Isolation Forest has a direct geometric representation.

In operational settings where resources are limited and accountability is expected, these characteristics have great significance.

In his study, Krzyszton performed experiments over two IoT-specific datasets, finding that the Random Forest method provides high levels of detection accuracy along with the ability to be interpreted by practitioners performing day-to-day incident response tasks. [13]. For many IoT deployments, classical methods are not a second-best compromise but the genuinely more appropriate choice.

1.3 Scope and Organisation of This Chapter

This chapter examines how traditional machine-learning methods can be used for detecting anomalies in traffic from IoT networks. The first part will provide information about the structure of IoT networks and what kinds of anomalies may occur in each IoT network, as well as provide a review of the major algorithm types and provide a review of the previous work done to benchmark these algorithms.

Subsequent sections will address some of the examples that illustrate how these techniques can be used practically and also present some of the challenges that are currently being faced in using these methods.

This chapter is meant to be read by individuals with some background in machine learning and/or computer networking. I will not be presenting any new algorithm(s), nor will I present my own empirical studies; instead, I will be using previously published material to give the reader a good foundation regarding what the state of the art in the field is.

Most of the references that I will refer to will have been published no later than year 2025, which I hope will provide a reasonable reflection of the current state of the literature on this subject.

2 Background and Conceptual Foundations

Prior to reviewing several types of detection methods, we should explain some of the characteristics of IoT networks that create uniqueness from the security aspect; how they are characterized by the types of anomalies existing in IoT environments; and why machine learning provides a natural basis for addressing such problems. This portion will define what those characteristics are, discuss some of these anomaly types, and then discuss why machine learning is appropriate.

2.1 IoT Network Architecture and Protocol Landscape

An IoT system is made up of three tiers: There are End Devices, which are any number of sensors or actuators, they collect data from or provide control commands for the physical world using Low Power and Low Data Rate protocols. These protocols can include ZigBee, Z-Wave, Bluetooth Low Energy (BLE), and LoRaWAN. Each protocol will have different characteristics related to range, throughput, and average energy consumption. [8]. There are two types of devices that connect directly to the internet and some that send their data to a local gateway before being sent on to the internet. The gateway must aggregate several data connections to the internet and serve as a translator between the different local protocols of devices and the many different protocols that the internet supports.

The middle tier is made up of the gateways, edge servers, and local network devices. Examples of devices in this tier include smart home hubs, gateways in industrial plants, and cellular IoT base stations. Because the gateway is the point where devices connect to the internet, and it has more processing power than the devices do, it's a good location for anomaly detection, since it'll be able to see lots of devices at once.

The upper tier is the cloud platform or backend where data is stored at scale, processed, and handled. The data traffic crossing from the gateway tier to the cloud crosses over the public internet and passes through many external risks, creating the many external risks to those devices.

This structure also creates a challenge in implementing a single static detection model across an entire deployment because there are so many kinds of device traffic (different methods of communicating between devices, multiple versions of TCP/IP protocols used to communicate between devices, and completely different data rates) that must all be

monitored using many different protocols with many different rates. [1]. Figure 1 illustrates this three-tier structure and the position of detection components within it.

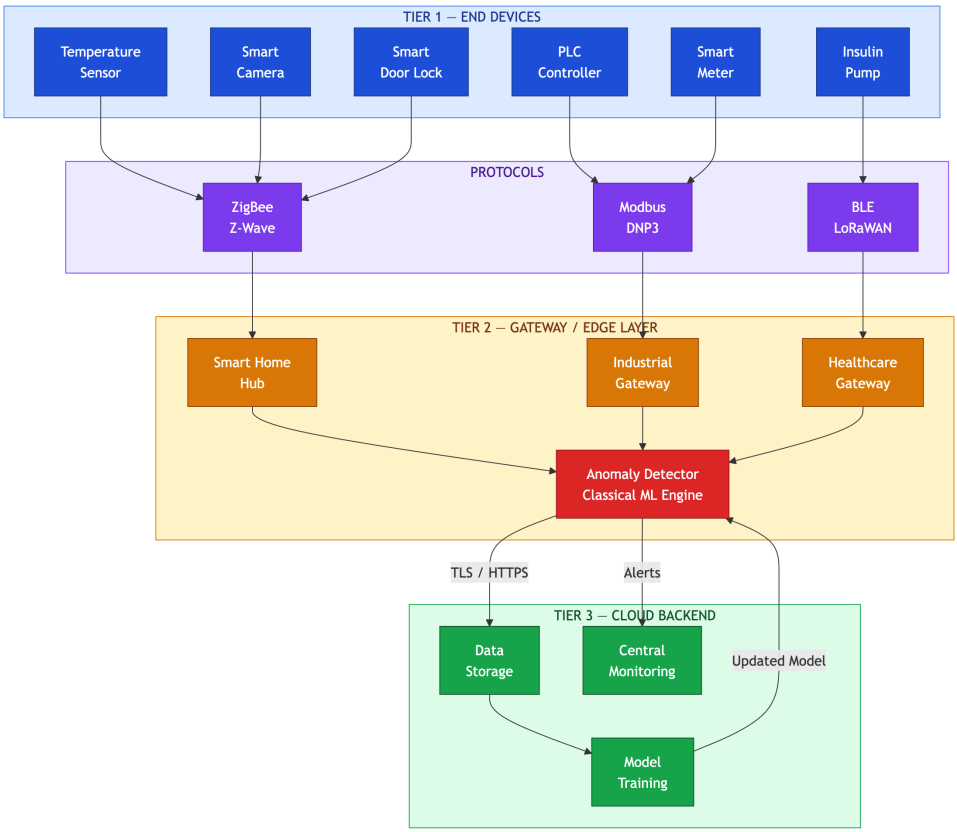

Figure 1: Three-tier IoT network architecture showing end devices, gateway/edge layer, and cloud backend, with representative communication protocols and detection placement at each tier. Adapted from Haider et al. [8].

2.2 Characterising Anomalies in IoT Traffic

Anomaly is used as a loose term within literature, this loose use of the term creates issues when comparing results between different papers. The work of Cook, Misirli and Fan identifies three different types of anomalies within IoT time-series anomaly detection, in alphabetical order they are; point anomalies (i.e. A single point that exists outside an expected range for normal), contextual anomalies (i.e. A value that is normal in one context but abnormal in another) and collective anomalies (i.e. A series of observations that together create an anomaly even though each individual observation would not by themselves) [5] - All three types of anomalies are present in IoT network traffic.For example, a single packet sent from a botnet infected device to an unknown destination would be a point anomaly. A second scenario would occur if this botnet infected device were to send packets with a volume matching the 'normal' level for this device except that the destination of the packet is to a command-and-control server instead of where it

would normally transmit traffic, creating a contextual anomaly as the volume is normal but the destination is abnormal. An example of a collective anomaly would be in the event of a coordinated distributed denial-of-service attack in which each participating device will only be sending a minimal amount of extra traffic than normal but combined will generate enough traffic from thousands of devices to overwhelm the target, as there would be no device-specific monitoring set up to detect the collective anomaly existing without the context of the entire attack [3].

Data collected about IoT devices are different than enterprise networks both positively and negatively when being identified as normal (i.e. established baseline/thresholds) and abnormal (i.e. attacks). The behaviour produced by individual IoT devices is usually highly predictable and consistent especially in their normal operating state. For example, Hasan reported that his machine learning classifiers (i.e. algorithms designed to identify patterns through the use of data sets) trained on IoT sensor data produced results greater than 99% accuracy when identifying normal (i.e. established) versus various types of attacks (e.g., DoS, scanning, etc.). Such high accuracy can be attributed to their predictable behaviour when in their normal state. [9].

2.3 Machine Learning as a Detection Framework

Machine Learning (ML) Techniques for Anomaly Detection treat the Anomaly Detection Problem as a Statistical Modelling Problem. In the case of Network Behaviour, for example, you are given a data set and your goal is to build a model that can determine whether or not a New Observation is Normal or Abnormal, or score the New Observation based on how distant it is likely to be from the Mean of all Normal Observations. The type of model that you can use depends on how much labelled data is available to Train on [11].For example, if you have both labelled examples of Normal and Attack Traffic, then you would apply a Supervised Classifier to build your model. Alternatively, if you only have, or have access to, labelled data for Normal Traffic because Attack Traffic is Rare, Novel or simply Unlabelled, then you would need to use Unsupervised or One-Class learning techniques to build your model. Semi-Supervised techniques use a large set of Unlabelled Normal Traffic data and a small set of labelled data and combine the two in order to use the benefits of Unsupervised Learning's Broad Coverage and Supervised Learning's Ability to Discriminate. Ahmad and Alsmadi have demonstrated in their systematic literature review of ML Techniques used for both Anomaly Detection and IoT Security that the majority of the published literature related to Anomaly Detection is based on Supervised Techniques because of the Availability of Benchmark Dataset Labelled for Use as a Reference Point and the High Rate of Accuracy Compared to Standard Evaluation Techniques [1].

Another element contributing to a classical model's effectiveness is that of feature

engineering. Traditional techniques usually involve practitioners using features from raw network packets or flows for training. Some examples of such features include summary statistics (e.g., mean, variance, and max) of packet size, distribution of inter-arrival times, count of each flag in protocol headers, ratio of inbound to outbound traffic, and entropy of destination IP addresses. Ayad et al. showed the hybrid use of feature selection methods (filter and wrapper) prior to classification improves both the detection accuracy and inference time for the BoT-IoT, TON-IoT, and CIC-DDoS2019 data sets, further supporting the need for proper feature engineering. choice [2].

3 Classical Machine Learning Techniques for Anomaly Detection

This section surveys the principal families of traditional machine learning algorithms that can be utilized in order to detect anomalous behavior within the IoT network.

The section will consist of supervised classification methods, as well as unsupervised and clustering approaches, along with ensemble methods that merge different approaches together.

3.1 Supervised Classification Methods

To systematically use support vector machines for an intrusion detection system will continue to be used as one of the earliest forms of machine learning algorithms. Regardless, they also continue to be highly competitive in today's IoT anomaly detection space. This process really comes down to finding the hyperplane that will best separate normal from anomalous traffic samples based on the large distance separating two classes in a high dimensional feature space. In addition, if normal and anomalous classes cannot be linearly separated in the original feature space, kernel tricks can be applied to allow the SVM to work with the transformed data space without actually having to calculate the transformation. The radial basis function (RBF) kernel has been used most frequently during IoT experiments involving intrusion-detection systems. [4].

The k-nearest neighbours algorithm is a method of classification that does not have to rely on an explicit training step. To classify a new observation, this algorithm finds k records from the training set that are similar to the new observation and takes the majority class label of those k records and makes that the label for the new observation. The concept of anomaly detection says that if a query point is surrounded by a predetermined number of neighbours (the letter k) that are classified as anomalous, this query point will be classified as anomalous; if, however the query point is surrounded by a majority of normal traffic, then it will be classified as normal. Research by Mohy-eddine has shown that by applying Principal Component Analysis (PCA) for feature selection to

a K-NN classifier created by combining both PCA and K-NN resulted in 99.99% accuracy when using the Bot-IoT dataset and reduced the time to perform a prediction from greater than 14 hours to less than 1 minute, effectively demonstrating how the selection of an appropriate feature set can change the K-NN algorithm from being impractically slow to being operationally feasible. [14].

The usage of supervised learning methods for IoT anomaly detection has made significant advances thanks to the invention of decision trees and their descendants, such as ensemble methods. Decision Trees separate regions in the feature space by using threshold rules for each of the features used to segregate the data. Random forests use bootstrap samples from the original training set to build a collection of decision trees and then use the aggregated predictions from these trees to reduce the variance in predictions from one tree alone. Gradient boosting builds new trees and corrects the errors from the previous tree, resulting in the best classification performance compared to conventional methods with structured tabular data. In research conducted by Hasan, the random forest model consistently produced better results than both logistic regression and decision tree baselines on IoT sensor attack datasets when evaluated using multiple metrics of performance (precision, recall, AUC) even though the raw accuracy (i.e., overall correct prediction rate) measures among the three methods were similar. [9].

A Gini Impurity Weighted Random Forest Feature Selection Method (GIWRF) by Disha and Waheed was suggested specifically for IoT intrusion detection. Their work showed that weighting the trees based on class distribution along with Gini impurity for selecting split points improved the F1 score over standard random forests for both the UNSW-NB15 dataset and the TON_IoT dataset. [7]. The pattern of this work is a part of a broader trend that has shown that most successful supervised IoT anomaly detection applications are those which use a carefully adapted version of an established algorithm, rather than a fundamentally different one, to the characteristics of their target dataset.

3.2 Unsupervised and One-Class Detection Methods

Unsupervised Anomaly Detection is the process of detecting anomalous events in unlabeled data samples. This type of anomaly detection uses an unsupervised approach, meaning that the model has been trained without the use of labels and has no known examples of attack traffic. Unsupervised anomaly detection is useful in IoT contexts where collecting and labelling attack traffic is challenging, but requires greater complexity from the underlying algorithm since the detector needs to discover what constitutes normal from the structure of the unlabeled data.

Isolation Forest is a type of model which has been specially developed to be used for detecting anomalous events, whereas a general classifier would not be able to do so as effectively. [16]. An ensemble of random binary trees, built using a method that relies

on repeatedly randomly selecting a random feature and randomly picking a split point on that feature, helps the isolation of anomalous data points compared to normal points. This is because most anomalous data points tend to be more sparse in nature, therefore making them easier to find and represent more spread out bases of above average distance from the other records in the data set.

The Anomalies, therefore, will have an Anomaly score associated with the number of split decisions (shown by the number of split decision branches traversed) to isolate this anomaly. Each point that will be traversed utilizing very few split decision branches will therefore have a high Anomaly score while those that will require numerous split decision branches will have an Anomaly score that is likely to be a normal record.

Zahoor additionally, used the TON_IoT dataset to evaluate the Performance of the Isolation forest and the One Class SVM algorithms by conducting an importance analysis based on random forests to ascertain the validity of the data produced through five-fold cross-validation as artifacts of a specific split of the data. [16]. Their analysis showed that OCSVM is likely to have the highest precision and recall, while the Isolation Forest produced competitive results with a much lower computational overhead when faced with adversarial label-flip poisoning attacks. OCSVM's "one-class training" adaptation of the standard SVM methodology allows for training using only examples of normal classes. An OCSVM learns an enclosed region or boundary in the feature space that is bounded by a hypersphere hyperplane encapsulating the majority of the training data from the normal classes. When the classifier is applied to new observations, these will either be classified as normal (inside the boundary) or as an anomaly (outside the boundary). There are two controls ν on how to balance the portion of training samples falling outside the boundary versus the size of the volume of the enclosed region, and these were tuned between the two studies based on the results of using (LIME) to produce one examples characteristics that present the largest contribution or significance toward the decision to classify an example as an anomaly [16].

DBSCAN (short for Density-Based Spatial Clustering of Applications with Noise) is one example of how density-based clustering is done. It defines clusters as dense regions of similar data, separated by regions of data that have little or no density (i.e., sparse). In other words, density-based clustering defines noisy points as not in any cluster, just as DBSCAN does.

In network traffic analysis, DBSCAN has been used to identify botnet command-and-control traffic, which tends to reside in small, very dense clusters away from the main body of general traffic.

DeMedeiros et al. found that density-based methods are particularly useful when performing offline analysis of historical captured traffic data for detecting anomalous behavior in IoT and sensor networks. The authors of the survey state that density-based methods are very effective when there is no prior knowledge about the number or shape

of anomaly clusters from which to start. [6].

3.3 Ensemble and Combined Approaches

An ensemble of machine learning algorithms combines a number of individual classifiers into one overall prediction. An ensemble of classifiers that performs majority vote among individual classifiers (the voting ensemble) is a simple class of ensemble. A meta-classifier is trained on the combined outputs of individual classifiers to learn to determine which classifier to rely upon in which scenarios (stacking, or stacked generalisation).

Using the Bot-IoT dataset to perform an experimental comparison to evaluate seven different types of machine learning algorithms for both binary (2 classes) and multi-class (greater than 2 classes) classification. [4]. The authors discovered that while Random Forests produced very high performance (99 percent accuracy) on binary-classification problems for HTTP DDoS attacks, k-Nearest Neighbors performed equally well at 99 percent on multi-class problems. The authors conclude that because different classifiers perform optimally on different formulations of the same base dataset, ensemble methods should be considered; e.g., combining classifiers or building combinations of multiple classifiers will produce a more complete view of the performance of a classification algorithm over the long term.

Using six different machine learning classifiers over the UNSW-NB15 dataset, Kayo used a dimensionality reduction technique (principal component analysis) to develop an ensemble approach that consistently provided predictions comparable to the best performing classifier alone and reduced the sensitivity of the prediction to hyperparameter tuning of the classifier. [10]. Their studies further indicate that if PCA-based feature reduction; performed prior to classification (rather than afterwards); reduces both training time and increases the ability of detection methods to maintain their accuracy.

In addition, evaluation of whether to use an individual or an ensemble approach can be performed by examining the following factors: availability of labels for training the individual detectors; computational resources available for detection; acceptable level of false positives; and whether the most important criteria in selecting a method are to detect only those attacks that have certain markers of prior attacks, or to also detect new attacks that do not have many previous attack markers.

4 Comparative Discussion

When comparing anomaly detection techniques, evaluative standards should be taken into consideration. In this section you will find detailed information about what types of benchmarks are typically used in the literature regarding anomaly detection, what types of metrics are employed to determine utility of performance, and what types of

computational limitations exist for their deployment in an IoT context?

4.1 Benchmark Datasets and Their Scope

A large part of the research on anomaly detection depends on the type of datasets used to evaluate results; there are also benchmarks that have been published regarding the characteristics of each set. Among these, the UNSW-NB15 dataset (which was developed by the Australian Centre for Cyber Security) has many types of classes for evaluating attacks (such as reconnaissance, exploits, denial of service, and command injection) as well as many combinations of them that will lead to various results for anomaly detection. For example, Kayode used UNSW-NB15 as her main evaluation set and found that the accuracy of the machine learning classifiers ranged from 99.9% to 99.97% for the Matthews Correlation Coefficient metric; however, these high values reflect the characteristics of this specific environment and do not necessarily predict how well the classifiers will perform in all IoT environments. [10]. Disha and Waheed also evaluated UNSW-NB15 versus TONIoT and noticed diverse classifier orderings between these datasets, thereby confirming findings from one benchmark do not equate to finding the same across both benchmarks. [7].

Real IoT devices were infected in a testbed area and then had all traffic captured to create the Bot-IoT data set. Mohy-eddine used Bot-IoT as a means of evaluating their K-NN intrusion detection method because the traffic characteristics specific to IoT made the data set more fitting for use with an IoT focused classifier than prior general-use network data sets. [14]. Koirala, Bista, and Ferreira validated the use of Bot-IoT with an independent dataset of botnet traffic generated from IoT devices in the healthcare sector and found a high degree of agreement between the two data sources regarding which features were the best predictors of classification. [12]. This sort of validation between different datasets is not widely found in the literature yet can provide more reliable indicators of generalisation than the sole evaluation using a single dataset.

Krzyszto carried out an evaluation of methods using IoT23 and CICIoT2023 methods, these datasets are more recently published and reflect current behaviour of devices and attack patterns as opposed to the UNSW-NB15 or BotIoT datasets. [13]. According to these findings, the quality of data preprocessing has a greater impact on detecting accuracy than the choice of algorithm for most of the methodologies examined in this study, which can have significant implications for teams that are building IoT detection pipelines.

4.2 Performance Metrics and Evaluation Trade-offs

The fraction of observations classified correctly (accuracy) is not an appropriate measure for detecting anomalies, because the amount of attack traffic is small compared to the

Table 1: Benchmark datasets commonly used for IoT anomaly detection evaluation, with key characteristics and limitations relevant to result interpretation. Based on [4, 7, 10, 13, 14, 16].

Dataset	**Year**	**Attack Categories**	**IoT-Native**	**Key Limitation**
UNSW-NB15	2015	Reconnaissance, Exploits, DoS, Command Injection, Shellcode, Worms	Partial	General network traffic; not generated from real IoT devices
Bot-IoT	2018	DDoS, DoS, Reconnaissance, Data Exfiltration	Yes	Limited device variety; heavily imbalanced class distribution
TON_IoT	2020	Scanning, DoS/DDoS, Ransomware, Backdoor, Injection, Password, XSS, MITM	Yes	Lab-generated environment may not reflect production deployment conditions
IoT-23	2020	Botnet C&C, DDoS, Port Scan, Coin Mining	Yes	Captures limited attack scenarios per traffic collection session
CICIoT-2023	2023	DDoS, DoS, Mirai variants, MQTT-based attacks, Reconnaissance	Yes	Relatively recent; fewer independent replications available in the literature

normal amount of traffic. A classifier that assigns all records as normal would achieve a high accuracy score in a data set where only 1% of the records contained attack traffic would not provide any level of security to anyone.

Other metrics that are useful are the detection/recovery rate (also referred to as recall or true positive rate), the false positive rate, precision, and the F1 score. [4, 9]. The ROC curve and the area under it summarize the trade-off between detection rate versus false positive rate across all possible classification thresholds, and are often used to compare classifiers on imbalanced datasets. While precision-recall curves provide complementary information, they are more informative when the class with the positive label is rare.

Ayad reported results for evaluating their hybrid feature selection method on all four metrics to ensure that the evaluation did not artificially exaggerate the performance of their system due to dataset imbalance, as well as used SMOTE oversampling to balance class counts between attack and normal classes before training on the dataset. [2].

The cross-validation technique chosen can affect how comparable results are across

research articles. Some research employs random train – test splitting which may cause such data from the training set to leak into the test set if records (such as sessions) exist in both sets. Zahoor used 5-fold cross-validation in their comparison of isolation forest and one-class SVM to ensure that the reported performance measures are indicative of consistent model performance and not simply due to a "good" data split. [16]. The lack of consistency in how evaluations are conducted continues to make it a challenge to compare one set of research papers with another in terms of their overall performance, when all three of the above factors vary from study to study.

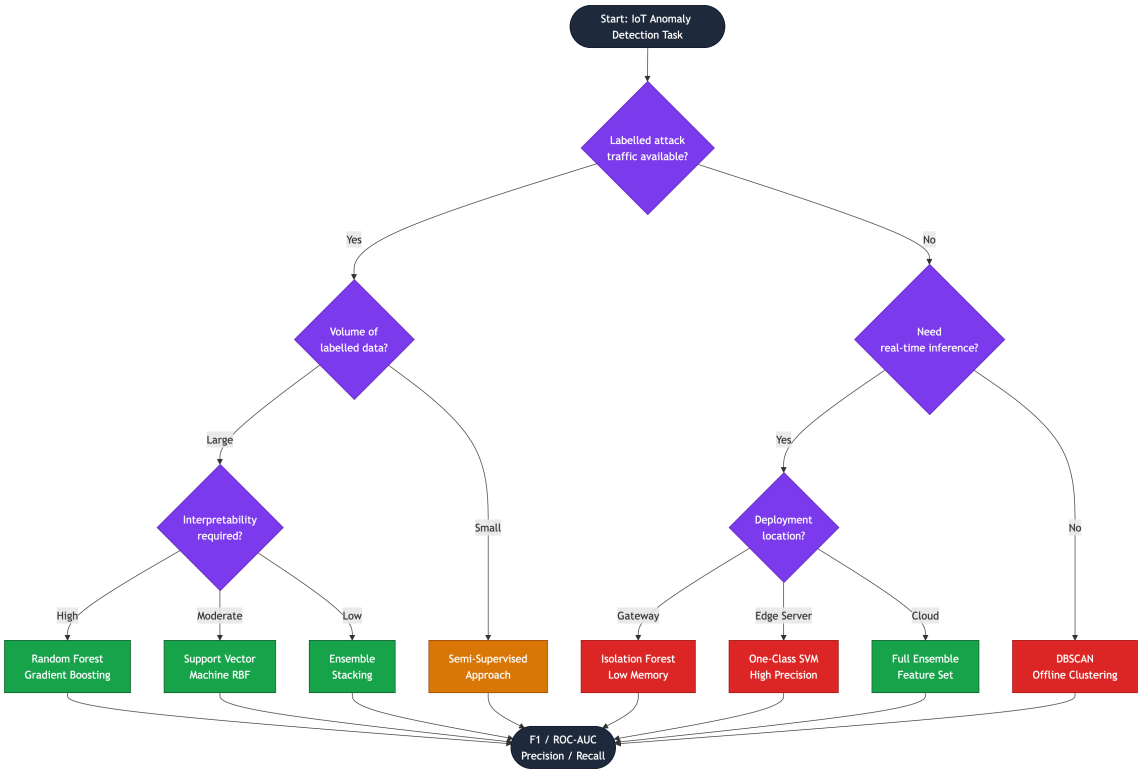

Figure 2: Decision flowchart for selecting classical machine learning algorithms for IoT anomaly detection. Selection logic and trade-offs between interpretability, data volume, and deployment location are synthesized from [4, 11, 13, 16].

4.3 Computational Feasibility Across Deployment Points

The resource constraints of IoT devices and gateways limit the complexity of models deployed in practice. A lowend IoT sensor usually runs a CPU with a frequency of 16MHz and has 32KB of RAM and 256KB of storage. A full random forest with hundreds of trees and thousands of features cannot run on this device. Therefore, detection must take place at the gateway/edge server/cloud, all of which have different latencies and availability. [8].

The use of gateway detection has become the most effective intermediate method. An investigations demonstrated that applying a method of feature

detection in order to reduce the input data from the full data set to just ten of the most informative features, will reduce the amount of time it takes to predict using the k-nearest neighbour (KNN) classifier from 50,000 plus seconds to 1 minute with only a minimal decrease in accuracy. [14]. The findings of this report indicate that the penalty for using fewer features rather than those that result in the best performance on benchmarks is likely to be less than previously thought and that there may not be a fundamental disconnect between having an acceptable level of performance on a benchmark compared to being able to meet both of those objectives (i.e., a feasible solution in terms of computational resources and an acceptable level of detection accuracy) as they are presented in the existing literature.

For cloud-based solutions, there is no limitation on resources available; however, there is also the consideration of delay that may be experienced when an analysis is performed remotely on the data in the cloud, making it unsuitable for applications that are time-sensitive.

In most industrial control systems, the detection and response to a network anomaly may need to occur in millisecond time frames to avoid any potential for physical damage.

Using cloud based analysis solutions may create several seconds of delay in detecting or responding to network anomalies.

Ultimately, any solution to developing a valid deployment strategy will depend on latency tolerances, applicable communication bandwidth, and the expected cost associated with missed detections versus false alarms for a given operational environment. Table 2 This table summarises the key properties of the methods discussed in the chapter.

Table 2: Comparison of classical machine learning methods for IoT anomaly detection across properties relevant to deployment. Performance estimates are indicative and vary with dataset and configuration; see [4, 10, 13] for detailed experimental benchmarks.

Method	**Learning Type**	**Needs Labels**	**Interpretability**	**IoT Suitability**
Random Forest	Supervised	Yes	High (feature importance)	Good at gateway
SVM	Supervised	Yes	Moderate	Good at gateway
KNN	Supervised	Yes	High	Limited (slow at scale)
Isolation Forest	Unsupervised	No	Moderate	Good; low memory
One-Class SVM	Unsupervised	No	Moderate	Moderate
DBSCAN	Unsupervised	No	Low	Best for offline analysis
Ensemble / Stacking	Mixed	Partial	Variable	Depends on components

5 Practical Insights and Use Cases

Algorithms for detecting anomalies may be successful when tested against established data sets but may not be automatically applicable in production environments.The following sections cover three ways of applying traditional machine learning (ML) methodologies to real-world applications through Internet of Things (IoT) devices (for example smart homes, industrial control systems and medical IoT).The approach taken for developing an ML based solution depends heavily on the specific use case involving an IoT device and different scenarios will come with unique challenges/considerations/requirements.

5.1 Smart Home and Consumer IoT Networks

A smart home features a variety of devices including voice assistants, thermostats, light controls, door locks, cameras, appliances, etc. All these various devices communicate with each other over the home's router and through local profile hubs using a mixture of Wi-Fi, Bluetooth, Zigbee, and ZWave.

Viewing this system from the standpoint of detecting anomalies is very difficult because there can be such a wide and varied range of normal behaviour in a smart home. Individuals who live in the same house will use devices differently; individual schedules change every year; and, devices change the way they work due to firmware updates with no indications that a malicious actor has compromised a device. [3].

The results of the different methods of anomaly detection were successful, demonstrating that both the random forest and decision tree classifiers trained from per-device flow statistics generated strong results in detecting potential intrusions into smart home networks. Hasan conducted this analysis using five different types of classifiers – logistic regression, SVM, decision tree, random forest, and neural network classifiers – to assess performance based on four different metrics (precision, recall, F1, and test accuracy). For testing accuracy, both random forest and decision tree classifiers achieved 99.4 percent accuracy, outperforming logistic regression and SVM classifiers on all other three performance metrics. [9]. In this case, the easiest to interpret tree-based algorithms also provide the highest levels of accuracy, creating a situation where the conventional tradeoff between accuracy and interpretability is eliminated. Koirala et al. used their research on botnet attack detection in a smart monitored environment as a basis for testing the Bot-IoT dataset, confirming that their results were valid even outside of the test devices and attack scenarios for which they generated their data [12]. The use of Pearson correlation coefficient and entropy as feature selection criteria before applying classification algorithms allows for good statistical measurement of feature importance and usefulness in guiding the feature engineering process without requiring substantial domain knowledge about which features are relevant.

5.2 Industrial Control Systems and Critical Infrastructure

The Industrial Internet of Things (IIoT) is comprised mostly of manufacturing facilities, power generation plants, water treatment facilities, and transportation systems. In these environments, devices known as operational technology (OT) are also located such as programmable logic controllers, remote terminus units, and sensors connected together via different industrial communication protocols such as Modbus, DNP3, or PROFINET. Anomalies detected within the IIoT carry significantly more risk than those found with a consumer based IoT; if a controller within either a power plant or a water treatment facility was compromised, it would not only have the potential to cause physical damage to the facility but also affect the safety of the general public depending upon that facility for safe energy or potable water [6]. The IIoT has a more regular behaviour pattern than what has traditionally been seen with the consumer IoT and this regularity works positively for the detection of anomalies. Controllers within the industrial sector communicate according to strict time schedules, to specific peer addresses and at fixed data message lengths. As a result, even small deviations from the standard of expected behaviour are flagged, as they would be considered suspicious in nature. The three types of anomalies (point anomaly, contextual anomaly, and collective anomaly) discussed by Cook, et al., can all occur in an IIoT environment; however, there is a heightened importance with regards to contextual anomalies as they indicate a device has been compromised and is relaying information to the attacker. An example of this would be if a controller was sending messages of normal length, at regular intervals, but sending them to an unexpected destination [5].

According to their analysis of current research trends, Rafique et al. noted that IIoT security is one of the circumstances wherein deep learning methodology's limitations are most pronounced, due to the necessity of understanding the functioning of a safety-critical system, the requirements for interpretable outputs would lead to the use of classical techniques rather than deep learning methodologies [15]. In order to take any type of physical intervention, operators of power generation facilities or water treatment plants would need to understand the reason(s) that the detection system raised an alarm, whereas an unexplainable output from a neural network lacks the necessary information to satisfy this requirement. Random forests will provide features importance scores that could be acted upon, for geometric anomaly scores, isolation forest will give a closer actionable score than would be achieved through the traditional method used for anomaly scoring or through the application of deep learning methodologies alone.

5.3 Healthcare IoT and Wearable Devices

The Internet of Things (IoT) in the medical field includes wearable technology like smartwatches and fitness trackers that have biometric sensors; devices that are implanted into

people such as insulin pumps, pacemakers, and cochlear implants; and monitoring devices that are connected to the hospital (such as ventilators, infusion pumps, and patient monitors). Anomalies within the network of devices in the Healthcare IoT will affect patient safety beyond just the data security issues associated with IoT devices in other industries. If an intruder has access to inject a command into an infusion pump or alter the data being sent to a glucose monitor, that could put someone in jeopardy of serious injury or death with these types of IoT devices [6]. The two methods recommended for Health IoT devices for anomaly detection in this type of IoT are the isolation forest and one-class SVM, due to the technical and ethical reasons why collecting real medical devices with labelled attack traffic is impractical. Zahoor et al. show that both methods provide effective anomaly detection in an unsupervised environment on the TON_IoT dataset; however, the one-class SVM performed with higher precision than isolation forest and required less computing time. [16]. In a patient monitoring setting where caregivers may ignore future alerts or disable monitoring due to false positive results, an appropriate default is one-class SVM due to its high precision characteristics. Cook, Misirli Fan have described one of the most actively researched areas for detecting time-series anomalies in healthcare wearables as being defined by the ability to establish a consistent baseline detecting deviations that result from network interference or device tampering. This is based on the regularity of physiological sensor signal characteristics from healthy individuals [5]. There are further challenges associated with hospital-grade medical devices; they frequently run proprietary operating systems (OS) that do not receive frequent (if at all) updates regarding security measures. Furthermore, these devices communicate via legacy healthcare protocols not designed with security in mind. The use of derived HL7 and DICOM protocol feature sets at network gateways offers an initial point of reference for classical classifiers that could be leveraged within security initiatives when there is limited time to wait on emerging solutions that are specifically developed for IoT in the healthcare industry.

6 Challenges and Open Issues

Methods outlined in this chapter produce outstanding performance on benchmark datasets under regulated evaluation conditions. Unfortunately, a number of long-term challenges hinder the conversion of these performance results directly into operational deployments. This section presents the most important of those challenges.

6.1 Data Scarcity and the Labelling Problem

One of the largest ongoing problems with the research into detecting anomalies in the Internet of Things (IoT) is that it is very difficult to acquire actual, realistic labels for

datasets. Attack data with labels can only be created by executing controlled experiments where an IoT device is attacked and then all of the traffic generated is collected or simulated with a simulation environment where the performance of the IoT device is expected to occur normally (not actually). If you were trying to gather the data from an actual production network, privacy issues would not only be raised, but you would also be out of compliance with national regulations Chatterjee and Ahmed did a survey of 64 published papers about detecting anomalies in IoT and obtained these three issues as being the most frequently cited: data and concept drift, integrating systems which have different types of sensors, and limited availability of true, ground truth labelled data [3]. Using a limited number of benchmark datasets to evaluate the various detection methods has created a negative feedback loop effect where the detection algorithms are essentially only optimized to work with the exact features and attack types which were included within the benchmark dataset. Thus, the performance numbers reported in most detection algorithms are inflated, with little to no actual improvement in the detection performance of those same algorithms when deployed against production network data. Ahmad and Alsmadi did a systematic review of the literature on machine learning methods for IoT security and concluded that the vast majority of the machine learning approaches developed for IoT security are evaluated using only one or two datasets, which significantly limits any confidence that can be established concerning their generalization [1].

By developing its own dataset based on the actual IoT botnet attacks, Koirala demonstrates how researchers can create a credible basis for their GI (Generalisation Identifiers) using a cross-validation approach between their data and existing publicly available datasets without waiting for additional datasets to become available. [12]. Another proposed avenue for generating synthetic attack traffic is through the use of synthetic data; specifically, generative models can be used to create realistic synthetic attack traffic for different types of devices that do not have existing datasets. However, evaluating the quality of this synthetic data can be very challenging because you would need access to real data in order to accomplish that – data that existed prior to creating the synthetic data set.

6.2 Hardware Constraints and Real-Time Deployment

In order to deploy trained models to devices located close to the Internet of Things (IoT), those models must be small enough to fit the available memory, fast enough to process traffic at line rate, and energy efficient enough that they will not impact the battery life of the gateway devices. These constraints limit many of the standard configurations of machine learning that have performed well when tried in server side experiments. According to Rafique et al., because of the limited resources found on IoT devices, complex detection systems cannot be directly hosted on resource constrained IoT devices. There-

fore, great majority of practical deployments look to either a gateway based or cloud based deployment for most installations [15]. Research on lightweight model variants has specifically targeted this mismatch. Mohy-eddine et al. indicated that if principal component analysis feature selection was performed prior to applying the K-NN classification algorithm, the prediction time was reduced approximately 50,000 times, while maintaining an accuracy of over 99 percent; thus showing that with proper preprocessing, extremely large efficiency gains are obtainable [14]. Ayad et al. also showed that with their hybrid filter-wrapper feature selection procedure, they were able to reduce the amount of time it took to classify to anywhere from .02 seconds to .15 seconds, depending upon the dataset and classifier combination used; demonstrating that there are really no shortage of classical methods that can meet the speed requirements of near-real-time IoT monitoring with classical methods [2].

Another challenge to consider when deploying classifiers is that classifiers training on traffic from one year ago will be less accurate over time as device firmware updates change how devices behave. Therefore, it is important for classifiers to keep their models current so that they can effectively predict traffic behavior. However, collecting and labelling enough new training data would not only be difficult, but it may also be nearly impossible in a real-world production environment. Incremental learning methods allow for a classifier's parameters to be updated as new training data become available rather than having to completely retrain its model each time there is a change to the environment; however, incremental learning methods should be designed so that they do not throw away previously learned norms in order to accept and incorporate new traffic patterns into their models.

6.3 Adversarial Robustness and Concept Drift

The field of AI/ML is regarding adversarial ML. Basically, what this means is that there are people out there who try to create inputs that will trick machine learning models into providing the "wrong" answer. There is no doubt that in the fields of computer Vision and Natural Language Processing, there has been plenty of previous research on these types of attacks. However, with regards to Network Anomaly Detection, there are also adversarial machine learning attacks, which are basically defined as input/traffic patterns that closely resemble "normal" (non-threatening/benign) behaviours but have been created with the intention of deceiving the network anomaly detector. Zahoor et al. recently performed an adversarial evaluation of their isolation forest (IF) and one-class SVM (OCSVM) models using simulated label-flip poisoning attacks on their training data set. Their results showed how resilient the models were to the effects of the poisoning attacks; i.e., depending on the amount of training data that was corrupted, and that OCSVM maintained an acceptable level of detection rate within moderate levels of poisoning,

while IF degraded significantly more than OCSVM under similar circumstances [16]. The concept of Concept Drift can simply be described as changes, whether gradual or abrupt, in the statistical distributions of all types of traffic over time. For example, as new devices are added to an IoT ("Internet of Things"), firmware versions get changed on devices connected to the IoT, and there are periodic seasonal and organisational changes in network usage patterns; the normal baseline that the anomaly detector was trained on may no longer match current network conditions. In their recent comprehensive review on ML Based Intrusion Detection Methods, Kikissagbe and Adda determined that concept drift is one of the major structural challenges faced by all deployed Intrusion Detection Systems. They highlighted that IoT environments experience an unprecedented rate of change in comparison to 'traditional' Enterprise Networks [11].

There are significant challenges created by the combination of adversarial robustness and concept drift to maintaining continuous detection systems through retraining while remaining vulnerable to adversarial poisoning attacks where an attacker intentionally injects adversarial samples into the training stream to adversely affect the updated model. The difficulty in managing this tension between being adaptable to change and being robust against adversarial attacks remains among the toughest problems in computer science that currently are not solved in practice, with no demonstrated production-level solutions to both issues.

7 Future Directions

In the previous section, the challenges were discussed that give an indication of some future directions for research in this area over coming years. Three of the most active and important areas will be represented in this section.

7.1 Federated and Distributed Learning for IoT Security

Federated Learning provides a way of training a shared model across multiple devices/organisations without exchanging any original input data. Each individual participant uses their own local dataset to perform their own model training, sharing only the model parameters or gradients with a central co-ordinator who aggregates these updates into a single, common model. Federated Learning can be used in an IoT AVD function to provide a mechanism for building AVD models across multiple different networks while mitigating the risk associated with the centralising of this data by using federated model updates instead of centralised models. According to Haider et al., Federated Learning represents one of the most promising future research directions for IoT AVD. They note that Federated Learning will help to solve both the data scarcity problem (by providing a means to aggregate learning from many deployments) and the data privacy issue (by keeping raw traffic

on-premises) at the same time [8]. In addition, they discuss that due to 6G's ultra-low latency and high level of connectivity, distributed AVD systems will be able to respond much more rapidly than traditional methods as the high-bandwidth and low-latency links from gateways to edge servers will significantly reduce the communication overhead of sending/receiving federated model updates.

The huge obstacles are still present. Many of the devices that will serve as gateways for the internet of things do not have enough processing power to allow the computations to perform their work. The amount of available bandwidth between an edge device and a central coordination point can be limited or intermittent. Despite these issues, there is a clear direction moving forward for the use of these technologies, with early experiments providing evidence that the performance of models built on multiple federated datasets by using an individual approach far exceeds that of a model developed using only the data from one local dataset.

7.2 Explainability and Trust in Deployed Detection Systems

An alarm system that alerts to some anomaly but does not provide an explanation of why it has triggered is not helpful and can make the job of the analyst impossible. For example, the analyst must be able to determine what about the flagged traffic is unusual in order to determine if it is a legitimate attack, a false positive created by benign device behavioural changes; or if the anomaly is due to a gap in the training data of the detection model. Ahmad and Alsmadi have identified interpretability as one of the major research gaps in IoT security machine learning work; indicating that the focus on accuracy in developing benchmark dataset models has not yielded models that can provide meaningful support to human decision-making in terms of effective use of the models. [1]. In their research, Zahoor et al. made a practical contribution in this direction by applying Local Interpretable Model-Agnostic Explanations (LIME) by generating per-observation explanations for their isolation forest and one-class SVM models, which indicated which features contributed significantly to classifying a particular network flow as anomalous [16]. This is not limited to only those two algorithms; there are many other types of classical machine-learning models that can be explained using LIME and/or similar post-hoc explanation methods, thereby providing a practical mechanism for improving the operational utility of deployed detection systems without needing to change the underlying model. In a related manner, Rafique et al. suggested that the widespread deployment of AI-based anomaly detection in healthcare and industrial control environments will depend on the ability for regulators and operators to be able to audit and understand the audit process for detection decisions [15]. The development of explanation frameworks for network anomaly detection is an area of ongoing research with high practical implications, particularly in contrast to the methodologies for generat-

ing explanations that have been developed for image classification and natural language processing.

7.3 Adaptive and Self-Updating Detection Pipelines

When static detection methods are established, they are not modified after deployment; so, therefore, as changes in the environment occur, these systems will soon fail to provide accurate detection. Adaptive models are the logical solution for this situation. When using adaptive models, the model will monitor its own behaviour and determine when to update as necessary based on the strength of drift detected in the underlying data set. To achieve this level of adaptivity in IoT systems, it is necessary to solve several highly-intertwined problems. Namely: detection of when an existing model has become outdated, determination if there are legitimate reasons to consider a change in model behaviour due to drift or an attempt to alter the baseline of detection due to an adversarial event, collection of sufficient new observations so that the model can be accurately updated, and deployment of the updated model (without interfering with current on-going detection activity).

According to the research by Cook, Misirli and Fan sampling time series anomalies in IoTs, online learning has been identified as a highly relevant method for these applications and represents the future direction for meeting these challenges [5].

When new observations are received, online learning will adjust the parameters of the model incrementally, which makes them suitable for high-throughput network monitoring on hardware that has bandwidth constraints. While typically possessing lower accuracy than batch-mode-trained ensembles when testing against static benchmarks, online learning will maintain adequate accuracy as drifting environments become unstable and rapidly degrade the performance of batch-mode-trained model(s).

According to Haider et al., edge computing will enable us to establish adaptive detection systems, since edge servers provide a greater extent of computational capacity than individual gateway devices and allow for more frequent updates of an intrusion detection model without needing to transmit traffic to the cloud [8].

According to Kikissagbe and Adda, the next generation of IoT intrusion detection systems will likely consist of hybrid approaches that combine supervised learning, unsupervised learning and deep learning techniques to create a new type of intrusion detection. Classical techniques will provide the efficient and interpretable base layer, while using learned representations to enhance the detection of new attack patterns [11].

In order to create a scalable human-in-the-loop feedback mechanism, it will be necessary to develop methods on a systems design level and a machine learning level to provide an evaluation of what constitutes a false positive and what constitutes a true positive.

8 Conclusion

The chapter examined the use of classical machine learning methods for detecting anomalies in IoT network traffic. Covered in the discussion are the foundational conceptual and architectural aspects of the problem; key algorithms and their properties; potential evaluation criteria; practical environments where these methods could be deployed; and the difficulties associated with not fully achieving these methods' goals within an operating environment.

Highlighting several key points will conclude the discussion. There is no single classical machine learning algorithm that consistently works the best for all types of IoT anomaly detection problems. Most of the time, supervised learning uses random forests and gradient-boosted trees as the best performing algorithms; however, when using unsupervised learning, either Isolation Forest or One-Class Support Vector Machine (SVM) are typically used due to not having access to labeled data for attacks. Research conducted by Churcher and colleagues and Hasan and colleagues have demonstrated that the ranking of various ML techniques used to classify data will depend on both how the ML problems are formulated and what dataset and evaluation metric used. This indicates that the specific characteristics of the environment they will be deployed in should determine which technique is optimal for the given application and not a general preference towards a specific algorithm [4, 9]. Another important consideration in addition to algorithm choice is the importance of feature engineering and feature selection. Research completed by Mohy-eddine and colleagues and Ayad and colleagues have demonstrated that if the feature space is reduced down to only a small number of highly relevant features, then both the accuracy and processing speed of the resultant classifier can be improved. This raises questions about the previous assumption that having more features will always improve detection performance [2, 14]. Ultimately, these results have real-world implications for teams who will deploy classical methods on resource-limited IoT gateways.

The benchmark datasets that researchers generally use for performance assessment have acknowledged shortcomings, therefore inferences drawn from a dataset may not be accurate representations of reality. Krzyszto'n et al. argue to expand access to larger, up-to-date and more heterogeneous publicly available datasets that reflect modern day device uses in the IoT anomaly detection literature. This argument is supported by the work of Koirala et al. where the results from one dataset are validated using another dataset [12, 13].

The difference between research output and application/output does not stem from the selection of the Algorithm but rather from a difference in distribution between training and applying, concept drift, physical limitations of hardware, and the adversarial nature of the Security Domain. To eliminate this gap will require coordinating work on Algorithm

Development, Systems Engineering, Data Set Curation, and Evaluation Methodology. It is not likely that the entire field will transition uniformly towards Deep Methods for all domains; therefore, Classical Methods will still exist side by side with Deep Methods, and Practitioners will select one or the other based not on results from Benchmarking By Test method. As such, it is clear that understanding what is known or not known about Classical methods is essential to anyone designing/evaluating an Internet of Things (IoT) Security System and is not simply a matter of historical perspective.

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